

Zero-Shot Conditional Molecule Generation with Latent Diffusion Models from Contrastive Pre-Trained Embeddings

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Motivation

Generating novel molecules that are active for given assays is a critical problem in drug discovery.

Challenge

Existing generative models often rely on direct conditioning with textual or molecular descriptors, or they require explicit training with guiding signals from the assay.

Idea

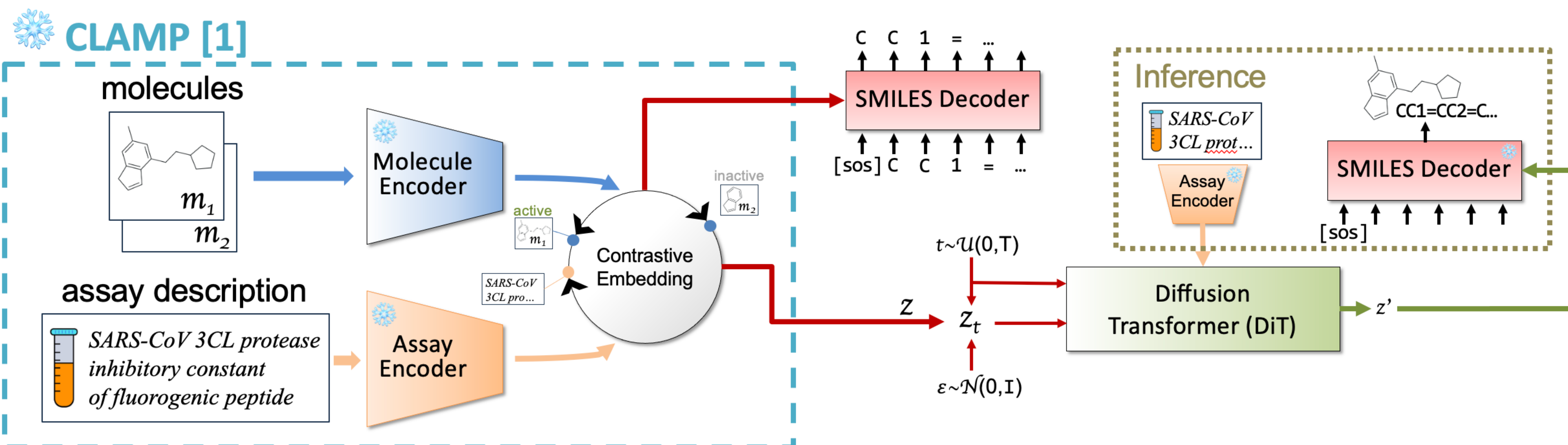
The proposed approach utilizes a pre-trained contrastive embedding space - designed to cluster active molecule-assay pairs - and employs a latent diffusion model to generate novel active molecules for a specified assay.

Introduction

- Drug discovery is a complex, lengthy (5–10 years), and costly (\$8B) process.
- Focus: Identify molecules targeting a specific disease or assay.
- Challenge: Only a small fraction of screened molecules show activity.
- Limitation: Existing models predict activity of known molecules but **cannot discover new structures**.
- Our Approach: Use contrastive embeddings to align active molecules with assays and **generate novel active molecules without explicit assay guidance during training**.

Contrastive Encoder [1]

- Molecule Encoder: Takes molecular fingerprints
- Assay Encoder: Takes text, describing the assay
- Contrastive Objective: **Active molecule–assay pairs are pulled together** in embedding space, while inactive pairs are pushed apart.



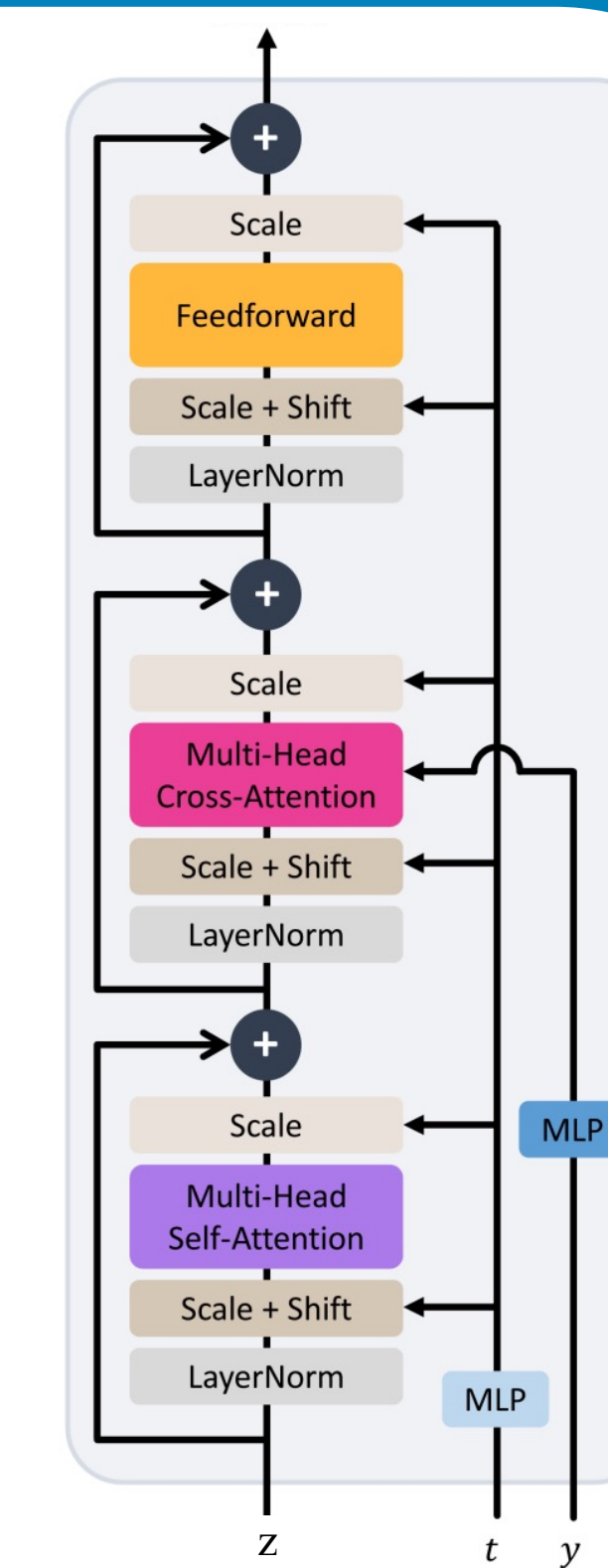
Latent Diffusion Model

- Contrastive embeddings as input from the CLAMP Model
- DiT blocks [2,3] for the forward/backward diffusion steps
- Train process without assay text guiding ($c = 0$)**

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I})$$

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t, c), \Sigma_\theta(x_t, t, c))$$

- During **inference**, we **guide the generation process** with an assay description.
- Guidance text is fed into the cross-attention layer



SMILES Decoder

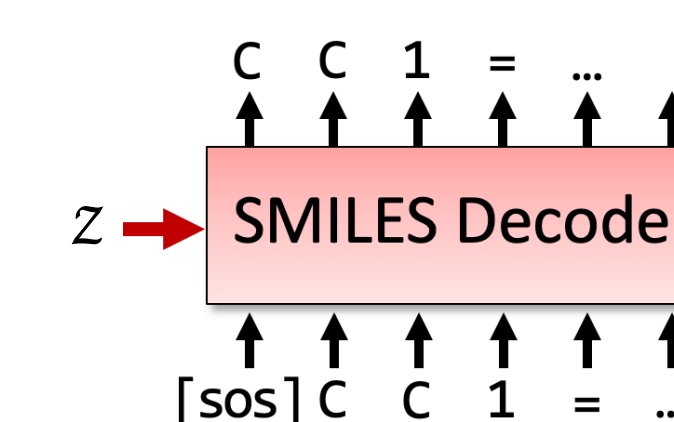
- Objective:** Convert a contrastive embedding into a valid SMILES sequence.
- Dataset:** PubChem: 10M molecules only
FS-Mol: 5k assays, 150M measurements; PubChem: 500k assays, 200M measurements

- Architecture:**
 - BART transformer
 - assay–molecule embedding is fed into the cross-attention of BART's decoder

- Sequence Generation**

- decoder predicts the next SMILES token x_t given prior tokens $x_{0:t-1}$ and the embedding z .
- Teacher forcing trains the model by providing ground-truth tokens at previous steps, minimizing the Loss

$$\mathcal{L}_{dec} = - \sum_{t=1}^T \log P(x_t | x_{0:t-1}, z)$$



Results

Model	Teacher Forc.	Free Decoding		
	Validation Accuracy	Tokenwise Accuracy	Sequence Accuracy	
LSTM	50%	19%	1%	
Masked Bert	80%	29%	9%	
BART	97%	68%	49%	
Model	Validity↑	Uniqueness↑	Novelty↑	FCD↓
DiT (unconditional)	92%	99.8%	100%	14.5

Future Work

- Condition on Assays**
 - conditional evaluation
 - add assay text during diffusion training
- Longer Training & Larger Bottleneck**
 - SMILES decoder to higher accuracy (above 90%)
 - dimension reduction for speed up and regularize
- Further Evaluations**
 - check generated molecules are indeed active
 - zero-shot performance
- Comparison with Baselines**
 - Implement other known generative models as baselines to compare performance.

References

(1) Seidl et. al (2023). ICML; (2) Peebles, Xie (2023). arXiv; (3) Chang, Ye (2024), arXiv